

Rifat Chowdhury

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EDUCATION

Brock University

Bachelor of Science, Computer Science

Sep 2019 – Apr 2025

St. Catharines, ON

EXPERIENCE

Promotions and Marketing Student Coordinator

Sep 2022 – April 2023, Sep 2023 – April 2025

Brock University Student Union

St. Catharines, ON

- Led initiatives to increase student engagement and awareness of Union services, events, and governance at Brock University
- Promoted a wide range of campus events to keep students informed and connected throughout the semester
- Adapted promotional strategies based on evolving student interests to maintain relevance and maximize engagement

Corridor Co-op Associate of Operations

Jan 2022 – Aug 2022, May 2023 – Aug 2023

Ministry of Transportation

St. Catharines, ON

- Tested and stress-tested internal systems to identify and document major bugs for the development team, improving system reliability and user accessibility
- Participated in project meetings to provide recommendations and flag potential system issues for ongoing online service improvements
- Supported deployment of an ArcGIS mapping tool used to track permit, inquiry, and pre-consultation files

PROJECTS

WeatherWise GitHub | Demo | *React, Node.js, Express, SQLite, Open-Meteo API, REST APIs*

2026

- Built a full-stack weather application with location search, browser geolocation, real-time conditions, and a responsive 5-day forecast interface using React, Express, and Open-Meteo APIs.
- Developed RESTful backend features for saving, updating, deleting, and exporting weather records in SQLite, with input validation, graceful error handling, and CSV/JSON data export support.

Weather Trend Forecasting GitHub | Demo | *Python, Time Series Forecasting, Data Visualization*

2026

- Built a Python pipeline to clean and analyze 161K+ global weather records, aggregating 831 days of observations and applying IQR-based outlier handling.
- Developed and evaluated short-term temperature and precipitation forecasts using lag and rolling-average features, with automated charts and MAE, RMSE, and MAPE reporting.

HR Analytics Dashboard GitHub | Tableau | *Power BI, Python, SQL, Excel, Data Cleaning*

2026

- Built an interactive HR analytics dashboard using Power BI and Tableau with executive KPI summaries to track headcount, attrition, hiring trends, and employee demographics.
- Cleaned and structured HR datasets using Python, SQL, and Excel, enabling detailed employee-level drilldowns by department, role, location, and employment status.

VibeMap GitHub | Demo | *Python, Time Series Forecasting, Data Visualization*

2026

- Built a full-stack Spotify music recommendation platform that generated personalized song suggestions by clustering tracks on mood and tempo; delivered the experience through a React web app, Flask REST API, and CLI.
- Developed an end-to-end ML pipeline with SQL, SQLite, Python, and scikit-learn for data cleaning, EDA, K-Means clustering, and similarity-based recommendations; managed delivery across 10 Agile sprints with 100+ completed tickets.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, C/C++, R, SQL (PostgreSQL, SQLite), HTML/CSS

Frameworks & Libraries: React, Node.js, Flask, FastAPI, pandas, NumPy

Developer Tools: GitHub, Jira, Miro, DBeaver, Docker (basic), Vercel, GCP, IntelliJ, Eclipse, Visual Studio

Data Skills: Data Cleaning, Exploratory Data Analysis (EDA), Dashboarding, Power BI, Tableau, QA/Regression Testing, Bug Reporting